



Vidhyadhan Charitable Trust

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Affiliated to, Veer Narmad South Gujarat University, Surat

Luxelle – Premium Fashion & Lifestyle

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Group No.	11	

Project Overview & Mission



- Designed for luxury fashion brands and boutique markets
- Delivers a seamless and engaging digital shopping experience
- Connects premium brands with modern digital customers
- Improves both user experience and backend management
- Simplifies online shopping while maintaining premium feel

Luxury - First Design Philosophy

Clean & Minimal

Highlight products with space and elegance.

High-Quality Visuals

Rich imagery and careful spacing.

Micro-Interactions

Smooth animations improve engagement.

Luxury Typography

Curated fonts & colors reflect exclusivity.

Problem Statements

Most e-commerce platforms lack premium & modern design.

Managing product variations (size, color, material) is complex

Poor UX leads to high cart abandonment rates.

Systems fail to scale with growing users and product data

No solution combining performance with design quality

Technology Stack — MERN

Mongo DB

Database

Flexible, Scalable, NoSQL storage.

Express.js

Backend

Secure & Efficient REST APIs.

Angular

Frontend

Dynamic, Component – based UI.

Node.js

Runtime

High Performance Server – Side Ops.

Frontend Development — Angular



- Structured, component-based architecture for maintainability
- Fast, dynamic page updates without full reload
- Client-side routing ensures smooth navigation across pages
- Modular design improves code scalability
- Delivers a fast, modern, and interactive user interface

UI Design & Styling

Tailwind CSS

Fast, Utility – First styling Framework.

Fully Responsive

Adapts Seamlessly to all screen size.

Consistent Layout

Spacing & Typography rules throughout.

Modern Trends

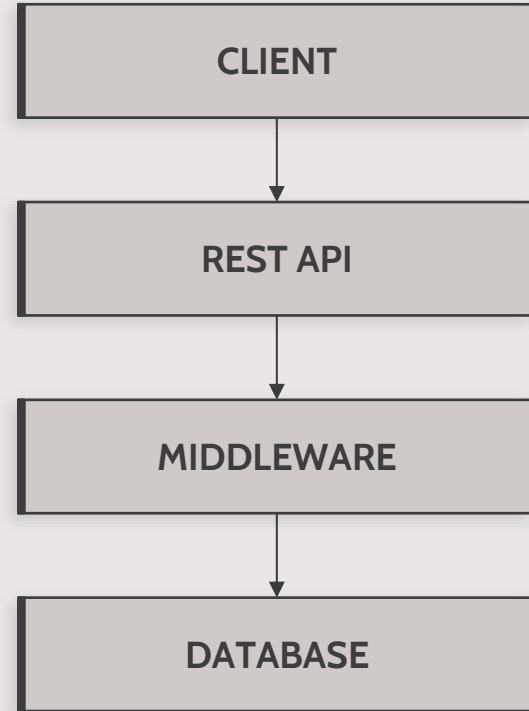
Gradients, Shadows, & Depth effects.

Luxury Aesthetic

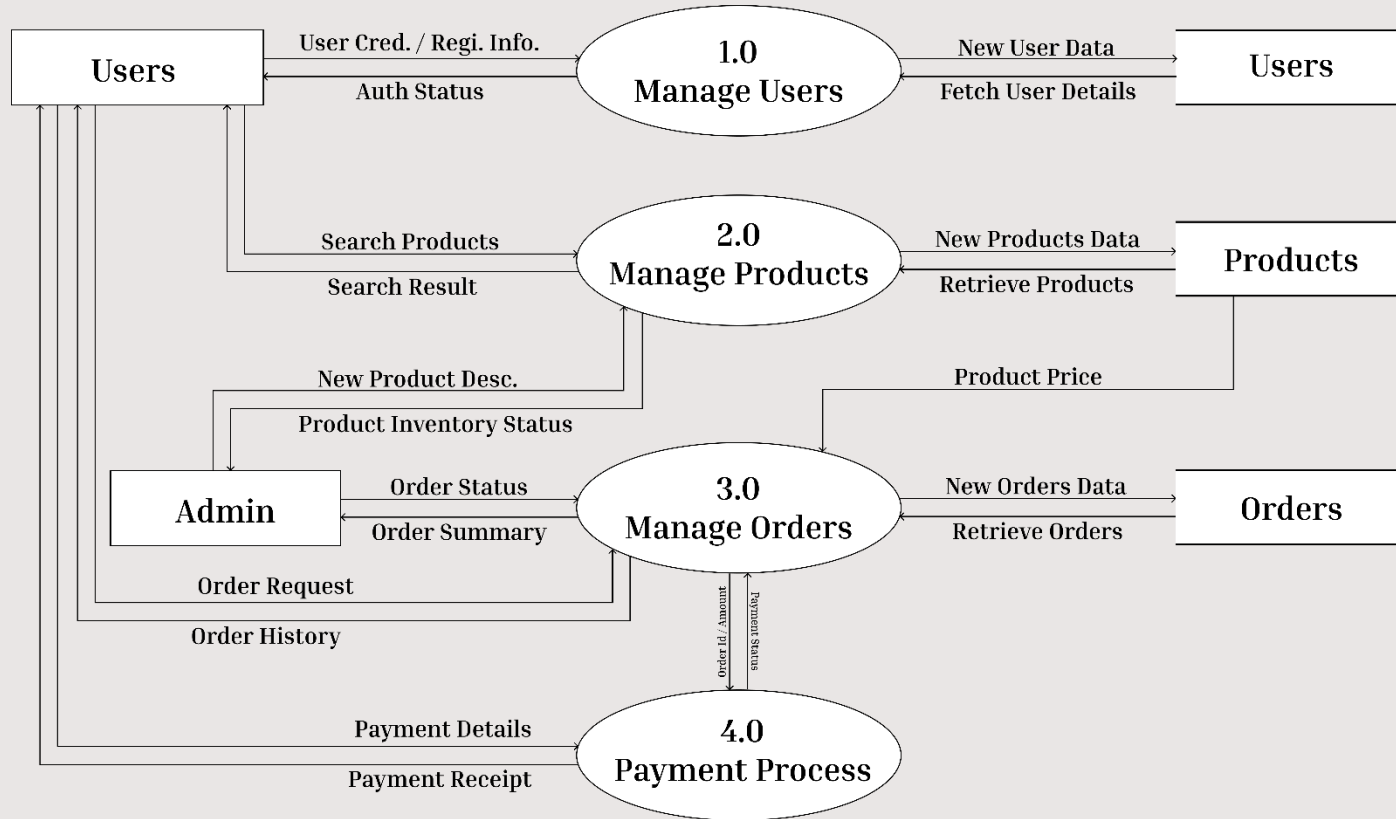
Elevated visual feel for Premium brands.

Backend & API System

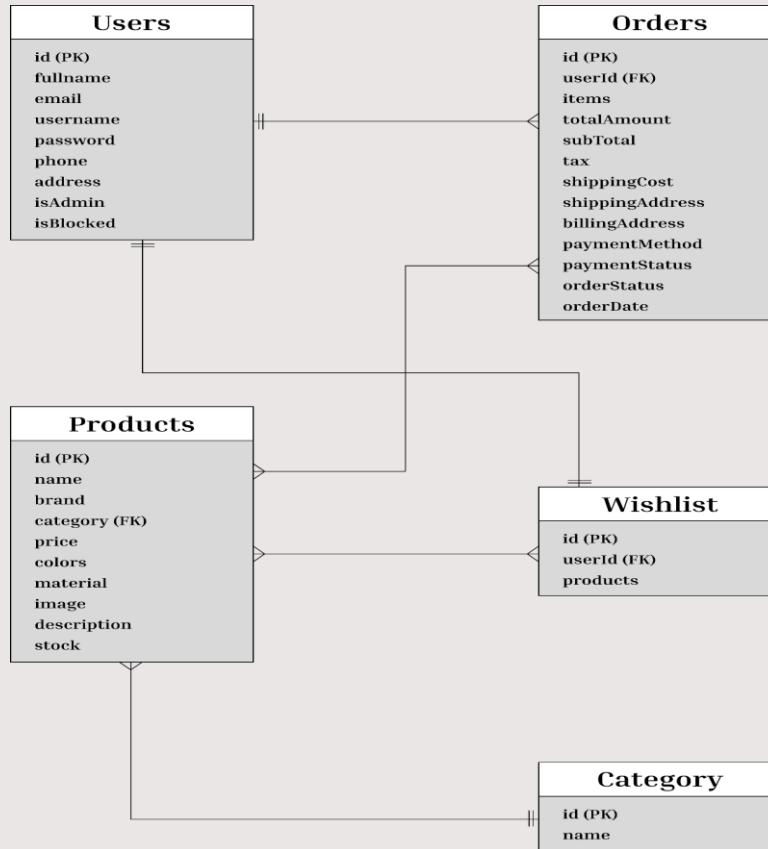
- RESTful APIs for communication between client and server
- Middleware handles authentication, logging, and validation
- JSON used for fast and efficient data transfer
- Business logic separated for better maintainability
- Backend ensures security, scalability, and performance



DFD LEVEL - 1

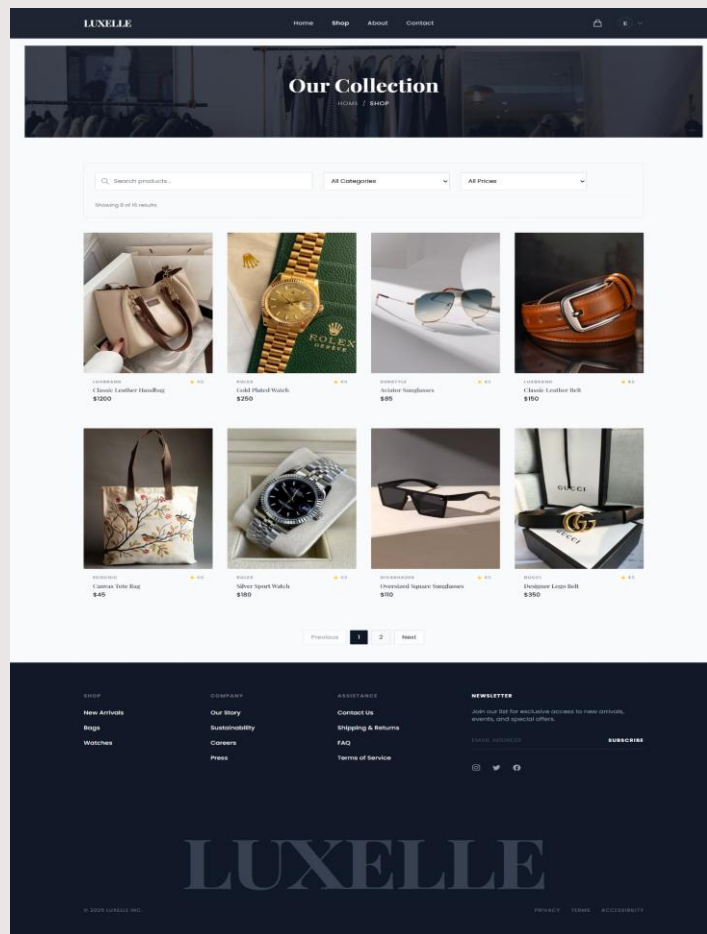


ER DIAGRAM



Product Discovery

- Browse products through categories and curated collections
- Detailed product info with high-quality images per listing
- Rich product details for informed purchase decisions
- Real-time stock availability displayed prominently
- Smooth, intuitive browsing experience throughout



Advanced Filtering



Price Range



Brand



Category



Color

- Multiple filters applicable simultaneously
- Interactive sliders for easy range selection
- Helps users quickly find desired products
- Improves overall shopping efficiency

Smart Search

- Search by product names, brands, or keywords
- Fast and accurate results delivered instantly
- Suggestions shown to improve search experience
- Search history helps revisit previous queries
- Ensures quick and easy product discovery

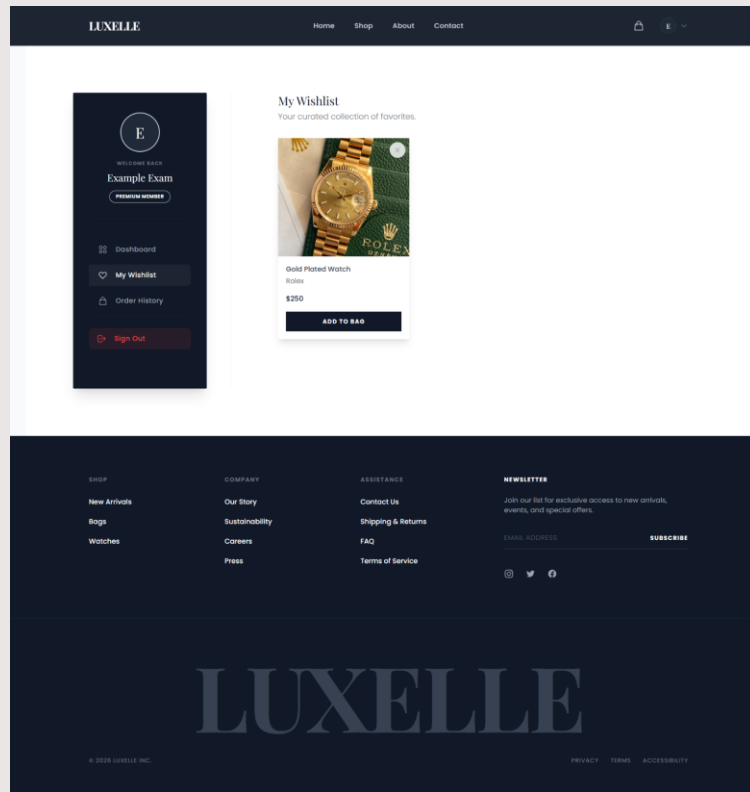
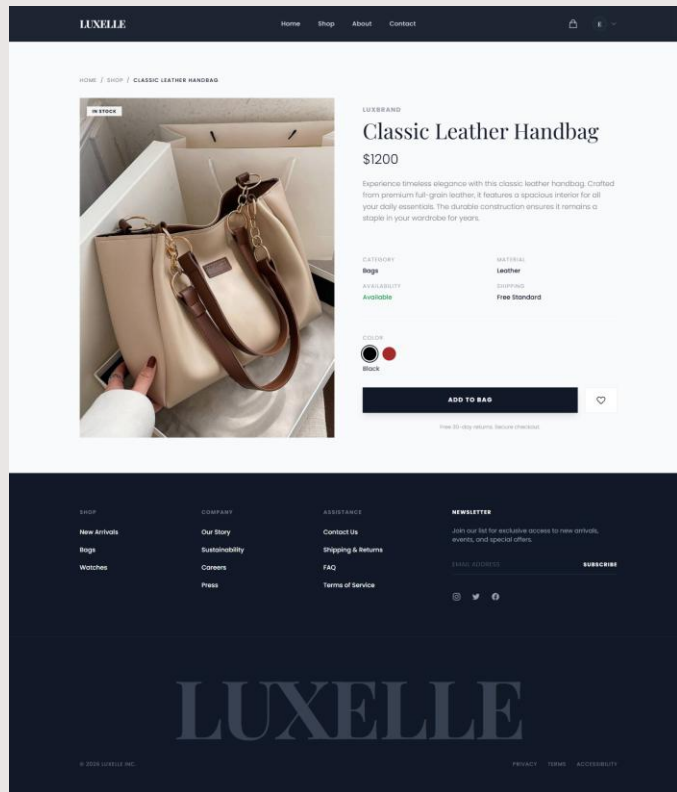
Search Luxury Fashion

Designer Bags

Silk Scaraves

Premium Watches

Product Details & Wishlist Page



Order Management

- View all past and current orders in one place
- Each order shows items, quantities, and pricing
- Order status updated in real-time
- Track purchases and complete transaction history

Placed

Processing

Shipped

Delivered

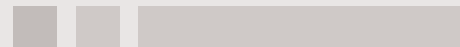
Wishlist System

Save products for future purchase with one click.

Wishlist persists across sessions and devices.

Move items to cart instantly from wishlist view.

Low-stock alerts notify users about saved items.



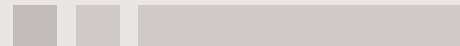
User Dashboard

Displays user profile and account information.

Quick access to orders, wishlist, and settings.

Easy update of personal details and preferences.

Central control panel for all user activities.



Admin Dashboard

Luxelle
Admin

MAIN MENU

Products

Users

Orders

Settings

Sign Out

Product Inventory

Manage your catalog and stock



Total Products
16
Items in catalog



Avg. Price
\$208.38
across all items



Categories
4
Active collections

+ Add Product

PRODUCT ITEM	CATEGORY	PRICE	STOCK	ACTIONS
 Classic Leather Handbag Experience timeless elegance w...	Bags	\$1,200.00	Stock: 28	 
 Gold Plated Watch Make a bold statement with this...	Watches	\$250.00	Stock: 29	 
 Aviator Sunglasses Soar high with these iconic avia...	Sunglasses	\$85.00	Stock: 100	 
 Classic Leather Belt Secure your style with this classi...	Belts	\$170.00	Stock: 80	 
 Canvas Tote Bag This versatile canvas tote bag is...	Bags	\$45.00	Stock: 100	 
 Silver Sport Watch Built for performance, this silver...	Watches	\$180.00	Stock: 44	 

Showing 1 - 6 of 16 products

< Page 1 >

Security Features

Password Encryption

Password Stored using bcrypt hashing.

JWT Authentication

Secure, Stateless User Sessions.

Input Sanitization

Prevents XSS & injection attacks.

Error Logging

System errors monitored & logged.

Database — Products Collection

Field	Type	Description
name	String	Product name & brand
category	String	Product category
price / discount	Number	Pricing details
stock	Number	Real-time inventory
images	Array	Product photo URLs
variations	Object	Size, color, material

Database — Users Collection

Field	Type	Description
name / email	String	User identity fields
password	String	Encrypted with bcrypt
role	String	Admin or Customer
wishlist	Array	Saved product IDs
addresses	Array	Multiple shipping addresses

Database — Orders Collection

Field	Type	Description
userId	ObjectId	Links order to user
products	Array	Items, quantity & price
status	String	Order lifecycle stage
shippingInfo	Object	Delivery address details
createdAt	Date	Timestamp of order

Future Scope

Advanced Payments

Integration of Stripe, Razor-Pay & more gateways.

Reviews & Ratings

User-generated product reviews and star ratings.

AI Recommendations

Personalized suggestions based on behavior.

Dark Mode

Full dark theme toggle for user preference.

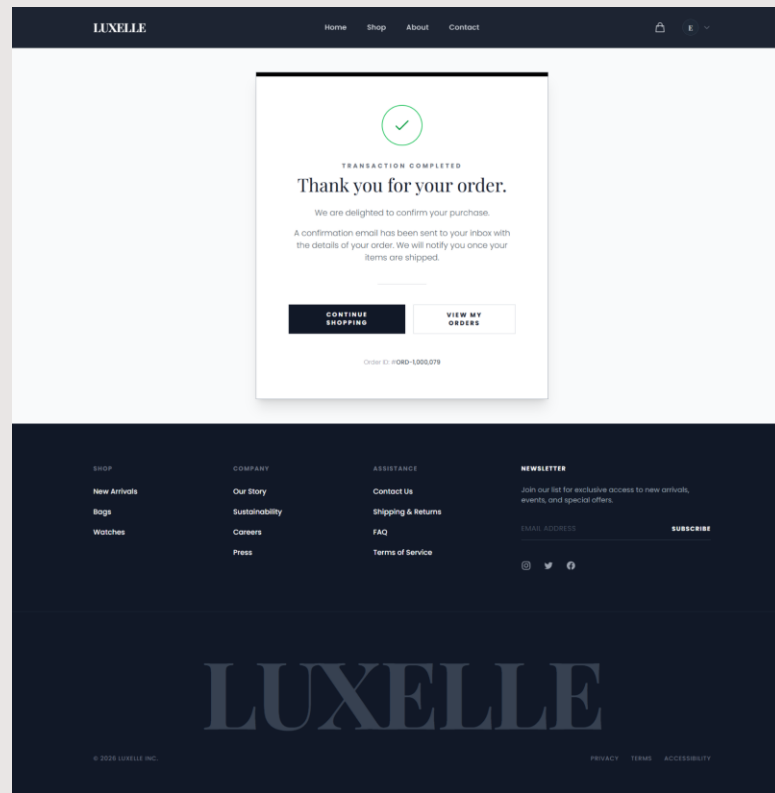
Scalability

Cloud-native deployment for global growth.



Conclusion

- Complete premium fashion e-commerce solution using modern web technologies
- Elegant UI design combined with a strong and scalable backend architecture
- Smooth, secure, and engaging shopping experience for every user
- Scalable platform capable of real-world business requirements
- A perfect blend of technology, performance, and luxury design



References

MongoDB

<https://www.mongodb.com/cloud/atlas>

Node.js

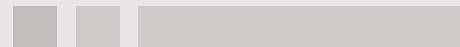
<https://nodejs.org/>

TailwindCSS

<https://tailwindcss.com/>

Angular

<https://angular.dev/>





Thank You!

Do you have any questions?

